



Prepared with Synoptic

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## Evidence and Argument Tables

# Rejected take-off and runway overrun, Tarlton Springs

The evidence, tests and reasoning behind the investigation's findings.

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Date of issue

09/10/2026

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## Contents

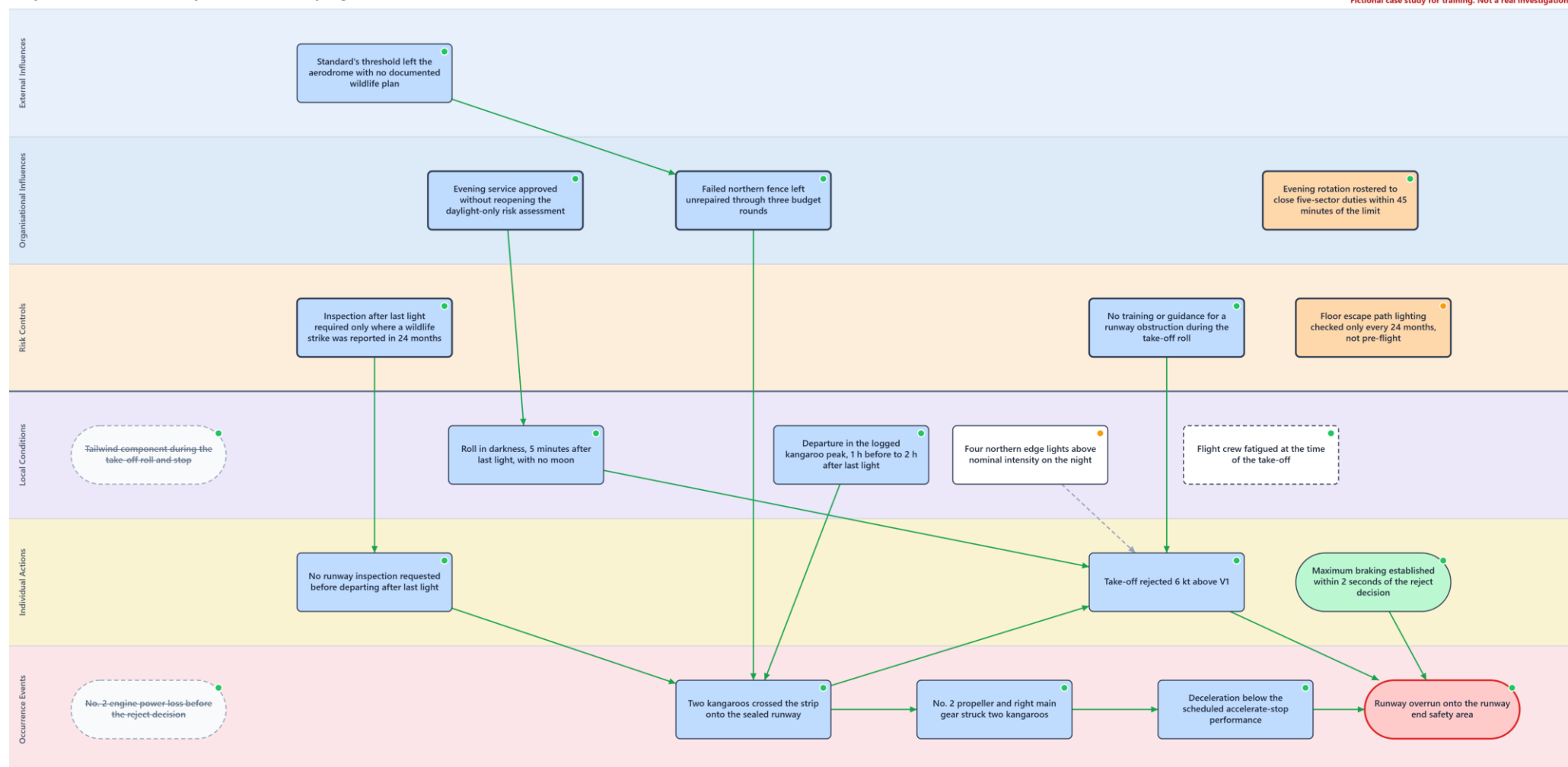
|                                                                                                   |    |
|---------------------------------------------------------------------------------------------------|----|
| AcciMap .....                                                                                     | 3  |
| How to read these tables .....                                                                    | 4  |
| External Influences .....                                                                         | 5  |
| Standard's threshold left the aerodrome with no documented wildlife plan .....                    | 5  |
| Organisational Influences .....                                                                   | 6  |
| Evening service approved without reopening the daylight-only risk assessment .....                | 6  |
| Failed northern fence left unrepaired through three budget rounds .....                           | 7  |
| Evening rotation rostered to close five-sector duties within 45 minutes of the limit .....        | 8  |
| Risk Controls .....                                                                               | 9  |
| Inspection after last light required only where a wildlife strike was reported in 24 months ..... | 9  |
| No training or guidance for a runway obstruction during the take-off roll .....                   | 10 |
| Floor escape path lighting checked only every 24 months, not pre-flight .....                     | 11 |
| Local Conditions .....                                                                            | 12 |
| Tailwind component during the take-off roll and stop .....                                        | 12 |
| Roll in darkness, 5 minutes after last light, with no moon .....                                  | 13 |
| Departure in the logged kangaroo peak, 1 h before to 2 h after last light .....                   | 14 |
| Four northern edge lights above nominal intensity on the night .....                              | 15 |
| Flight crew fatigued at the time of the take-off .....                                            | 16 |
| Individual Actions .....                                                                          | 17 |
| No runway inspection requested before departing after last light .....                            | 17 |
| Take-off rejected 6 kt above V1 .....                                                             | 18 |
| Maximum braking established within 2 seconds of the reject decision .....                         | 19 |
| Occurrence Events .....                                                                           | 20 |
| No. 2 engine power loss before the reject decision .....                                          | 20 |
| Two kangaroos crossed the strip onto the sealed runway .....                                      | 21 |
| No. 2 propeller and right main gear struck two kangaroos .....                                    | 22 |
| Deceleration below the scheduled accelerate-stop performance .....                                | 23 |
| Runway overrun onto the runway end safety area .....                                              | 24 |

Fictional case study for training. Not a real investigation.

# AcciMap

## Rejected take-off and runway overrun, Tarlton Springs

Rendered 08/10/2026, 2:34:20 pm  
Fictional case study for training. Not a real investigation.



Fictional case study for training. Not a real investigation.

## How to read these tables

Each table is one object from the analysis map. An object is a factor: an event (something that happened) or a condition (a state of affairs that existed at the time). Each factor is worded as a proposition, a statement that is either true or not, which the three tests then examine.

The tests run in order. Existence asks whether the factor existed or occurred. Influence asks whether it influenced this occurrence. Importance asks whether it is worth addressing regardless. A later test only means something once the earlier one is settled: Influence is tested once Existence has passed, and Importance where Influence failed. A test not yet reached reads Not tested.

The shaded row opens each test with its result (Pass, Fail, Inconclusive or Not tested) and the confidence behind it. Confidence follows the probability scale: virtually certain, extremely likely, very likely, likely, more likely than not, about as likely as not, unlikely, very unlikely, extremely unlikely and exceptionally unlikely, with the percentage in brackets. High confidence passes the test, low confidence fails it, and the middle is Inconclusive.

Under the result, Argument For sets out the case that the factor, as worded, is true for that test, and Argument Against the case that it is not. Both are always read against the factor as worded, never against the result, so a failed test still shows its case for. Reasoning says which side carried and why.

Nil means nothing was recorded on that side. A test argued both ways shows a case on each side, and the reasoning says which one carried. A test with nothing recorded shows only its result row: no argument was written for it, so its result is not supported by this record.

## External Influences

### Standard's threshold left the aerodrome with no documented wildlife plan

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> AS-139 10.4.2 requires a documented wildlife hazard management plan only where the aerodrome is used by aeroplanes above 27,000 kg MTOW or serves more than 350,000 passenger movements a year. Tarlton Springs' heaviest scheduled aeroplane is 22,900 kg and it handled 61,400 passenger movements in 2025, so neither limb applies. Under 10.4.3 its measures need not be documented at all. The aerodrome accordingly held no plan, no hazard assessment, no dispersal programme and no record of measures beyond the single pre-first-arrival inspection at Aerodrome Manual 6.3 and a sighting log that nobody reviews. Confirmed against the certified aerodrome register and the aerodrome operator's own compliance file; the aerodrome is not in breach.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> The standard's text, the aerodrome's traffic figures and the absence of any plan are documentary and not in dispute, and nothing in the record suggests a plan would have been held had the threshold not excused it.</p>                                                                                                                                                                                                                                                   |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Pass, likely (70%)</b>             |
| <p><b>Argument For:</b> With no plan, nothing owned the aerodrome's wildlife hazard as a hazard. The failed northern fence sat in the works register as an ordinary maintenance item competing for council money, and was deferred three times with the entry 'defer, unfunded, review next cycle' and no hazard assessment behind it. A documented plan brings a hazard assessment, an owner and an audit against both; under one, the fence is a recorded wildlife control whose failure has to be answered for, not a deferrable works item.</p> <p><b>Argument Against:</b> 10.4.1 still obliged the aerodrome operator to take reasonable measures, and a documented plan is not the only way to repair a fence or pass a sighting log to an airline. The council's deferrals were budget decisions, and there is no evidence a documented plan would have been funded any better. A plan is paper; the fence is money.</p> <p><b>Reasoning:</b> Argued. Accepted at 70% on the balance of the evidence: the deferrals were made with no hazard assessment in front of the council, which is the thing a plan would have put there, but the Against is real and a plan does not of itself fund a fence. Below the comfort of the other influence findings, and the report should say so rather than present it as settled.</p> |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Not tested</b>                     |

## Organisational Influences

### Evening service approved without reopening the daylight-only risk assessment

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> RRA-041 issue 3 (11/2022) is the current route risk assessment for Tarlton Springs. It carries the operating restriction 'daylight operations only pending review of night lighting and diversion fuel policy'. CMR-2025-118 (approved by the Head of Flying Operations on 22/05/2025) introduced a scheduled evening rotation from 14/07/2025, the first company operation into Tarlton Springs after last light. It assessed the change as 'schedule change only', recorded 'no new hazard identified', and answered 'No' to route risk assessment review required. RRA-041 was not reopened and no issue 4 exists. The Safety Department's records show no involvement beyond receipt of the approved record. SB512 is scheduled off blocks at 17:20 local; on 18/06/2026 it left on schedule and began its roll at 17:30, after the end of civil twilight at 17:25.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Virtually certain. The proposition tested is documentary and nothing contradicts it: CMR-2025-118 is a current controlled copy that records the assessment and the 'No' against route risk assessment review required, RRA-041 issue 3 is the current issue and carries the daylight restriction, and the document register holds no issue 4. The approver has left the company and cannot be interviewed, so why the change was assessed that way is unexplained, and that uncertainty belongs to the Influence test and to what the report may say about intent, not to whether the approval and the non-review happened.</p> |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Pass, likely (85%)</b>             |
| <p><b>Argument For:</b> The restriction, honoured, keeps every company departure from Tarlton Springs in daylight. The change put a company aircraft into the aerodrome after last light in explicit contradiction of it, through a process that asked whether the change was new rather than whether the assessment still held, and SB512 rolled in darkness on its schedule, not because of a delay.</p> <p><b>Argument Against:</b> the restriction names night lighting and diversion fuel, not wildlife. A review that reopened RRA-041 might have addressed those two items and approved the evening service with lighting and fuel conditions, leaving the departure after last light exactly as it was. The approver has left the company and cannot be asked what was in front of him, so what a review would have concluded is inference.</p> <p><b>Reasoning:</b> Argued. Accepted at 85%: the restriction is daylight only, and lifting it would have required a review to assess a night operation at an aerodrome with a single runway, no lighting redundancy and no alternate within 200 nm at night. The Against is the case where a review did that and approved the service anyway; it is possible, and it is the less likely of the two. Not certain, because the approver cannot be asked.</p>                                                                                                                                                                                                                                                                          |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Not tested</b>                     |

## Failed northern fence left unrepaired through three budget rounds

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> Works register item WR-221, raised 04/2023: four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers, with animals observed passing through, replacement estimated at \$41,000. Deferred at the 2023, 2024 and 2025 budget rounds, each entry reading 'defer, unfunded, review next cycle'. WR-263 (02/2026) records the same section deteriorating further. The investigator's site inspection on 23/06/2026 found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through the fence at the 1,050 m mark, 138 m before the point at which the first carcass was recovered.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Documented in the aerodrome operator's own register and confirmed on site.</p>                                                                                                                                                                                                                                                                                                                                                                                                    |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Pass, likely (80%)</b>             |
| <p><b>Argument For:</b> The failed section is the physical route by which the animals reached the movement area. The pad through the fence at 1,050 m and the carcasses at 1,188 m and 1,221 m are on the same side of the runway, and the register has recorded animals passing through this section since the day the item was raised. Repaired in 2023, the 700 m to 1,300 m section would have been intact in 06/2026 and this route closed.</p> <p><b>Argument Against:</b> WR-221 recommends replacement without stating a specification, and nothing in the record shows that the fence as replaced would exclude kangaroos, which can clear a stock fence. The pad shows these two came through the gap; it does not show that an intact fence would have stopped them, and the sighting log does not record where any of its 41 kangaroos came onto the aerodrome.</p> <p><b>Reasoning:</b> Argued. Accepted at 80%: an intact fence closes the one route the physical evidence shows these animals used, and the register's own entry ties the failure to animals passing through. Held below the 90% recorded before the Against was stated, because whether the replacement fence would exclude kangaroos is not in the evidence either way.</p> |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Not tested</b>                     |

## Evening rotation rostered to close five-sector duties within 45 minutes of the limit

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> The rostering system extract for 14/07/2025 to 18/06/2026 shows the evening rotation (SB511 Port Carrick to Tarlton Springs, SB512 Tarlton Springs to Port Carrick) operated on 289 days, each time rostered as the fourth and fifth sectors of a Port Carrick-based duty signing on between 06:00 and 07:00 and planned at between 12 hours 45 minutes and 13 hours 15 minutes, against the 13 hour 30 minute maximum for five sectors from those sign-on times. On six of those days a member of the rostered crew reported unfit and the rotation went to a crew already on duty, extending that crew's day from three sectors to five. 18/06/2026 was the sixth: sign-on 06:30, the take-off 11 hours in, a planned duty of 13 hours.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> The roster extract, the duty record and the departure times are documentary.</p>                                                                                                                                                                                                                                                                                                                                        |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Fail, unlikely (25%)</b>           |
| <p><b>Argument For:</b> The roster puts the last sector of the longest duty in the pattern into an aerodrome with no alternate within 200 nm at night and limited engineering support (RRA-041), and when a crew member reports unfit it hands that sector to a crew whose day has just been extended. On 18/06/2026 that put a crew who described themselves as tired into a night departure 11 hours into their duty.</p> <p><b>Argument Against:</b> the FDR gives 0.6 seconds from decision to power levers at flight idle and 2 seconds to maximum braking, at the fast end of the fleet's own distribution and well inside normal; there is no error of omission on the roll; and the one decision a long duty could plausibly have affected, not requesting a runway inspection, was never available to be affected, because no document prompted either crew member to consider it. The roster cannot be found to have caused the omission of a step nobody was asked to take, and whether the crew were fatigued at all is itself unsettled.</p> <p><b>Reasoning:</b> Failed on the evidence, not dismissed. The specific mechanisms the roster would have had to work through in this occurrence are each contradicted by recorded data.</p> |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Pass, likely (85%)</b>             |
| <p><b>Argument For:</b> It failed Influence for this occurrence and remains a finding worth acting on. Every operation of the evening rotation in eleven months has closed a five-sector duty planned to within 45 minutes of the maximum, at an aerodrome RRA-041 records as having no alternate within 200 nm at night and limited engineering support. When a crew member reports unfit, the rotation goes to a crew already on duty, and that has extended a crew's day from three sectors to five on the morning of operation six times in the period. Nothing about that changes because it did not bite here.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> A standing exposure built into the roster rather than a one-day event; it is there on every operation of the rotation and sharpest each time a crew member reports unfit.</p>                                                                                                                                                                                                                                                                                                                                                                                        |                                       |

## Risk Controls

### Inspection after last light required only where a wildlife strike was reported in 24 months

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> Operations Manual Part A 8.14.2 requires the pilot in command to request a runway inspection before the first departure after last light at any aerodrome listed in Appendix 8A, and not to commence the roll until the result is passed. 8.14.3 requires nothing at other aerodromes. Appendix 8A lists an aerodrome where a wildlife strike involving a company aircraft has been reported there in the preceding 24 months. Nine aerodromes are listed; Tarlton Springs never has been. No provision anywhere in Part A asks a crew to report a sighting, and no provision admits an aerodrome to the list on any other evidence. The list was last reviewed on 03/03/2025, four months before the evening service began.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> The manual text and the appendix listing are documentary.</p> |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Pass, very likely (90%)</b>        |
| <p><b>Argument For:</b> This is the operator's own defence against exactly this occurrence, and it was unarmed. Eleven sightings of kangaroos on the sealed surface at this aerodrome could not put it on the list, because they were not strikes and were not reported to the company.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Had the requirement applied at Tarlton Springs, an inspection would have been requested and a pass required before the roll, and the aerodrome's own staff disperse animals when they find them. The 10% allows that an inspection completed some minutes before departure does not guarantee a clear runway at the roll.</p>                                                                                                                                                                                              |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Not tested</b>                     |

## No training or guidance for a runway obstruction during the take-off roll

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Pass, extremely likely (95%)</b> |
| <p><b>Argument For:</b> TRG-SYL-07 module 4's four scripted scenarios are engine failure at 90 kt, engine fire at 100 kt, configuration warning at 85 kt, and engine failure at V1 plus 5 kt continued. All four are aircraft system events; none presents an obstruction, an animal or any external hazard on the runway, at any speed. No crew has been presented with an external obstruction scenario in the five training cycles on file, covering 214 crew in the current cycle alone. Part B 4.6 matches: 4.6.3 lists the reject items to V1, 4.6.4 states the take-off shall be continued at or above V1, and the section contains no guidance on an obstruction seen on the runway at any speed and no amplification of 4.6.4 for a case in which continuing is itself hazardous. Both the pilot in command and the first officer stated independently that every stop or go scenario they have been given has been an engine, and that nobody has ever put something on the runway in front of them in training.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Syllabus and manual are documentary; the 5% reflects that scenario records beyond five cycles were not available.</p> |                                     |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Pass, likely (75%)</b>           |
| <p><b>Argument For:</b> The captain had about a second and nothing behind him. With no scenario and no guidance, the decision he made was the first time he had considered the question at all, which is what training and procedure exist to prevent.</p> <p><b>Argument Against:</b> the captain stated plainly that he knew the rule and departed from it deliberately, not because he was unaware of it. Training him on obstructions would not have changed what he knew; he did not reject because he was untrained, he rejected because he judged the alternative worse. On that reading the gap is a finding about the fleet, not a cause of this reject.</p> <p><b>Reasoning:</b> Argued. Accepted at 75%: an unrehearsed decision made cold in about a second is not the same decision as one made against a briefed and practised frame, and the report should carry both readings rather than assert the first.</p>                                                                                                                                                                                                                                                                               |                                     |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Not tested</b>                   |

## Floor escape path lighting checked only every 24 months, not pre-flight

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> The pre-flight emergency lighting test on this type illuminates the ceiling emergency lights and the exit signs; it neither illuminates nor monitors the floor proximity escape path segments, whose only check is a dedicated functional test at the maintenance programme's 24-month interval. On VH-SBQ that gap held a live defect. Raised by the first officer, who noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation, the functional check on 24/06/2026 found two of the six floor proximity segments aft of row 16 on the left side dark on battery power with the system armed, traced to a failed common ballast. The last dedicated functional test had been 11 months before the occurrence, inside the interval, so the defect was undetectable by the crew or by a line engineer and could have stayed so for up to 13 more months.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> The test's coverage and the programme interval are documentary, and the VH-SBQ defect was found on a functional check and traced to a specific failed component.</p> |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Fail, unlikely (10%)</b>           |
| <p><b>Argument For:</b> Between functional tests the state of the escape path lighting on any aircraft of the type is unknown, and on VH-SBQ it was partly dark with nothing available to the crew to tell them.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> No evacuation was commanded, the cabin lights were on throughout, and the disembarkation took 4 minutes 8 seconds without incident. There is no path by which the monitoring gap, or the defect it hid, affected the overrun or its outcome.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Pass, very likely (90%)</b>        |
| <p><b>Argument For:</b> It did not influence this occurrence and it is still a finding. The disembarkation was unhurried, by the forward airstair, with the main cabin lights on. Change one thing, such as a fire indication, smoke or a night evacuation over the wing, and two dark segments in the aft cabin are the difference between an orderly exit and a blocked one. The gap is the monitoring: a 24-month interval on a system the daily test cannot see means each aircraft's true state is unknown until its next functional test, and whether the same ballast part is fitted across the fleet is still with engineering.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> A gap in the monitoring of an escape-path defence, which has already hidden one latent defect on this aircraft and can hide the same on any other of the type until its next functional test.</p>                                                                                                                                                                                                                                                                   |                                       |

## Local Conditions

### Tailwind component during the take-off roll and stop

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Fail, very unlikely (5%)</b> |
| <p><b>Argument For:</b> Hypothesis tested because a stop that runs 62 m past the end invites a question about the wind. It fails. The automatic weather station's 06:30, 07:00 and 08:00 UTC observations give 090 at 7, 090 at 6 and 090 at 5 kt, a headwind component of 5 to 7 kt on runway 09. Recorded FDR wind during the roll gives a headwind component of 4 to 6 kt throughout, with no tailwind component at any point. The crew's performance entry used a 5 kt headwind, which matches. There is no evidence of a tailwind at any stage of the roll or the stop, from any source.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Two independent sources, the aerodrome's automatic observations and the recorded aircraft data, agree that the component was a headwind throughout.</p> |                                 |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Not tested</b>               |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Not tested</b>               |

## Roll in darkness, 5 minutes after last light, with no moon

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> The roll began at 17:30 local (07:30:02 UTC on the FDR), on schedule, 32 minutes after sunset and 5 minutes after the end of civil twilight, with moonrise not until 22:41 and the moon 14% illuminated. The aerodrome was in full darkness with no natural ambient light on the strip and no town lighting within sight; the only lighting was the runway edge lighting the crew had switched on through the pilot-activated lighting at 07:21:50. The investigator's site visit at the same time of day on 23/06/2026 confirmed that the unlit graded strip either side of the sealed surface is not visible from the centreline.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Astronomical data and the recorded time of the roll; not in dispute.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Pass, likely (75%)</b>             |
| <p><b>Argument For:</b> In this light the strip either side of the seal cannot be seen, so the first a crew can see of an animal crossing it is the animal on the lit surface. The first officer's words on the CVR after the stop were that they 'came out of the dark on the left of the strip', and she told the investigation that outside the lights it was black and they 'didn't appear, they arrived'. In daylight, two animals crossing the graded strip from the northern fence would have been in view before they reached the seal, earlier in the roll.</p> <p><b>Argument Against:</b> the first officer did see them, at around 200 m, once they were on the seal, and how long they had been on the strip before that is unknown; animals that came onto the strip only seconds before reaching the seal would have been seen late in daylight too. The captain was heads-in on the engine page after V1, which is where the procedure puts him, so in daylight the sighting would still have rested on the pilot flying, whose attention belongs on the centreline.</p> <p><b>Reasoning:</b> Argued. Accepted at 75%: the site visit shows the strip cannot be seen from the centreline in this light, and the first officer's account is of animals arriving out of the dark, not of animals she was slow to see. Held below the other influence findings because the animals' time on the strip before the seal is not known.</p> |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Not tested</b>                     |

## Departure in the logged kangaroo peak, 1 h before to 2 h after last light

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Pass, extremely likely (95%)</b> |
| <p><b>Argument For:</b> The aerodrome's wildlife sighting log for 01/01/2025 to 18/06/2026 holds 46 entries, 41 of them kangaroos on or adjacent to the movement area. 38 of those 41 are timed between one hour before and two hours after last light, and eleven record animals on the sealed surface of runway 09/27. On 18/06/2026 last light was 17:25 local, so that window ran from 16:25 to 19:25; the roll began at 17:30. The captain's account matches the log: he has seen kangaroos on the strip at Tarlton Springs four or five times, always in the evening and never in daytime.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> The window and the time of the roll are documentary; the 95% reflects that the sighting log is a paper register whose completeness cannot be verified.</p> |                                     |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Pass, likely (80%)</b>           |
| <p><b>Argument For:</b> Kangaroos are on this aerodrome's movement area at this hour and seldom at others: 38 of the 41 entries fall in the window. Two animals crossing the runway at 17:30 is the pattern the aerodrome's own log describes, not a chance encounter, and a departure well outside the window would probably have met no animals on the runway.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> The log puts almost all recorded kangaroo movement in this window and the occurrence inside it. Held at 80% because the log records sightings rather than presence, and aerodrome staff go off duty at 18:30, part way through the window, so the pattern may partly reflect when there was somebody there to see.</p>                                                                     |                                     |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Not tested</b>                   |

## Four northern edge lights above nominal intensity on the night

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Pass, likely (80%)</b>                         |
| <p><b>Argument For:</b> A photometric check on 25/06/2026, seven days after the occurrence, measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal intensity for stage 3, in the section the animals crossed; the remainder were within tolerance. The runway lighting is pilot-activated: keyed from the CTAF, it runs for 30 minutes at the single intensity set at the aerodrome's lighting controller, which was found at stage 3, the setting recorded at commissioning. Crews cannot vary it. The works register and electrical log record no work on the lighting or its controller between 18/06/2026 and the check. The aerodrome's maintenance programme checks lamp serviceability but never measures intensity.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Likely rather than certain. The measurement was made seven days after the occurrence, not on the night, and lamp output can change; but the setting is fixed, nothing was done to the lighting in between, and nothing suggests the four lights were different on the night, so they are believed to have been at the measured intensity then, though nobody measured them at the time.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                   |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Inconclusive, about as likely as not (45%)</b> |
| <p><b>Argument For:</b> The four bright lights stand in the section the animals crossed. The captain recalled the lighting as 'brighter than I'd want', a corridor of light with everything outside it black, so that anything off the seal could not be seen coming, and said that with pilot-activated lighting there was nobody to ask to turn it down. Brighter edge lights deepen the contrast between the lit corridor and the dark strip beside it, which is where the animals came from.</p> <p><b>Argument Against:</b> the first officer, who actually saw the animals, described the lighting as normal and said she saw them completely clearly once they were on the sealed surface, at around 200 m. She also said plainly that with two daylight landings to compare against she is not well placed to judge whether it was unusual. The captain was heads-in on the engine page after V1, which is where the procedure puts him, so his non-detection has an explanation that does not require the lighting at all. The strip either side of the seal cannot be seen from the centreline in darkness whatever the edge lights are doing, and nothing establishes that a 131% edge light, rather than the darkness itself, delayed the detection of anything on or beside the runway.</p> <p><b>Reasoning:</b> Inconclusive at 45%, and left inconclusive rather than dropped or promoted. The intensity is the settled part; the claim that it delayed detection, and so the reject, is the unsettled part, and no evidence available reaches it. The accounts of the lighting disagree, the only witness to the animals saw them clearly on the seal, darkness alone explains why nothing off the seal was seen, and no study links a 131% edge light to the detection of an animal. The report owes the reader this reasoning rather than silence.</p> |                                                   |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Not tested</b>                                 |

## Flight crew fatigued at the time of the take-off

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| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Inconclusive, about as likely as not (50%)</b> |
| <p><b>Argument For:</b> the take-off was 11 hours into the duty, on the fifth sector, at 17:30 local after a 06:30 sign-on, and the day had been extended that morning from three sectors to five when the crew picked up the evening rotation from a crew member who reported unfit. Both crew described themselves in interview as tired at that point, and the captain said that if he had been asked to fly another sector he would have said no.</p> <p><b>Argument Against:</b> both crew described the tiredness as ordinary and not impairing. The captain said he was not struggling and that 'for the two seconds that mattered, I was there'; the first officer said it was a normal five-sector day, she was not fighting it, and she had had a proper break in Port Carrick in the middle of the duty. The duty was inside the company's limits, planned at 13 hours against a 13 hour 30 maximum, after two rest days for the captain and one for the first officer. The recorded performance shows none of the usual signs: the power levers reached flight idle 0.6 seconds after the decision and maximum braking was established at 2 seconds, at the fast end of the fleet's distribution, the callouts were made on time, and there is no error of omission anywhere on the roll.</p> <p><b>Reasoning:</b> Inconclusive at 50%, and left there rather than dropped or promoted. 'Tired' after 11 hours is what both crew said, and it is not the same as fatigued to a degree that degrades performance; the recorded performance argues against that, and no other evidence reaches it. Settling it would need a detailed fatigue assessment, with sleep history and the duty pattern before 18/06, which was judged not required: even had Existence passed, Influence would fail, because every mechanism fatigue would have had to work through here is contradicted by recorded data, and the one decision it could plausibly have touched, not requesting an inspection, was never prompted by any document.</p> |                                                   |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Not tested</b>                                 |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Not tested</b>                                 |

## Individual Actions

### No runway inspection requested before departing after last light

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> Neither the CVR nor the aerodrome's CTAF recording contains any request for a runway inspection, any offer of one, or any mention of wildlife at any point before departure, and the captain said wildlife was not discussed with anyone during the turnaround. The aerodrome's daily inspection log for 18/06/2026 records one inspection, at 07:35 local before the first arrival, result 'movement area clear', with none afterwards and none requested. Aerodrome Manual 6.3 provides for inspections outside the scheduled time on the request of the reporting officer or a pilot in command; the reporting officer was on duty at the terminal during the turnaround and confirmed one would have been carried out if asked. SB512 was the only departure after last light that day. Both crew stated they did not consider it; the captain had expected Tarlton Springs to be on the Appendix 8A list and was surprised to find it was not.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Established by two independent recordings, the crew's own accounts and the aerodrome's own log.</p> |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Pass, likely (85%)</b>             |
| <p><b>Argument For:</b> The one action that would have put a vehicle on the runway in the 20 minutes before the roll was available, costs nothing, and was never on either crew member's mind because no document put it there.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> An inspection would have covered the 1,050 m to 1,250 m section, where the fence pad and both carcasses were found, and aerodrome staff disperse animals when they find them. Discounted from certainty because an inspection some minutes before the roll cannot guarantee the runway stays clear.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Not tested</b>                     |

## Take-off rejected 6 kt above V1

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> The FDR records V1 called at 112 kt at 07:30:29 and both power levers retarded to flight idle from 07:30:32 at 118 kt, three seconds and six knots past V1. Part B 4.6.4 states the take-off shall be continued at or above V1. The CVR records the whole decision in one exchange: the first officer's sighting call at 07:30:31, 'Stopping, I have control' at 07:30:32. The captain's own account, in interview and in his air safety report, is that he judged rotating into two large animals would leave him airborne at night with unassessable damage over country with no usable option, and that he took the risk whose end he could see. He states plainly that it was not by the book.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Recorded data and recorded audio, corroborated by the captain's own account.</p> |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Pass, extremely likely (95%)</b>   |
| <p><b>Argument For:</b> Rejecting at 118 kt rather than continuing is the decision that put the aircraft into a stop it did not have the runway for. The scheduled accelerate-stop distance of 1,780 m assumes the reject begins at or before V1; begun at 1,152 m, the stop had 695 m of sealed surface left.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> The margin between ASDR and ASDA was 67 m. Beginning the stop six knots and three seconds late consumes that margin and more, before the reverse asymmetry is counted.</p>                                                                                                                                                                                                                                                                                                                   |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Not tested</b>                     |

## Maximum braking established within 2 seconds of the reject decision

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> Power levers reached flight idle 0.6 seconds after the decision and maximum braking was established at 07:30:34, 2 seconds after it, against a recorded fleet mean of 3.2 seconds from decision to maximum braking. Anti-skid was active on all four wheels throughout and brake energy reached 84% of the certified maximum without a brake fire. Reverse was selected on No. 1 within 4 seconds; No. 2 was unavailable because its propeller had been destroyed in the strike. The first officer made the speed callouts and the runway-end call. The aircraft left the seal at 21 kt, stopped 62 m into a 90 m RESA, upright, on the extended centreline, and was disembarked without an evacuation.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Recorded data throughout.</p> |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Pass, very likely (90%)</b>        |
| <p><b>Argument For:</b> The overrun was 62 m into a prepared surface at low speed rather than a departure from the RESA at speed. The margin between those two outcomes is the braking established in 2 seconds rather than the fleet's 3.2.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Recorded as a decrease in risk. A stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m, close to the far edge of the RESA.</p>                                                                                                                                                                                                                                                                                                                                  |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Not tested</b>                     |

## Occurrence Events

### No. 2 engine power loss before the reject decision

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Fail, very unlikely (5%)</b> |
| <p><b>Argument For:</b> Hypothesis tested because a reject at 118 kt is most commonly an engine event. It fails. The FDR records both engines at 98.4% and 98.5% torque, within 0.1% of each other, from power set at 07:30:02 until the power levers were retarded at 07:30:32, with fuel flow, ITT and propeller RPM on both engines inside normal scatter throughout. No caution, warning or exceedance is recorded at any point before the reject. The CVR contains no engine-related call: the captain's first words are 'Stopping, I have control' one second after the first officer's sighting call. The borescope of both engines found no pre-existing damage. The No. 2 propeller damage and the RPM decay both begin at 07:30:33, after the reject and at the moment of impact.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Recorded parameters, recorded audio and the physical inspection agree. The No. 2 loss is a consequence of the strike, not a cause of the reject, and the times separate the two.</p> |                                 |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Not tested</b>               |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Not tested</b>               |

## Two kangaroos crossed the strip onto the sealed runway

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| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> Two adult kangaroos were recovered from runway 09 at 1,188 m and 1,221 m from the threshold, on the northern half. The first officer saw them from the flight deck crossing from the northern side to the south, estimating a range of around 200 m at first sight, and called them at 07:30:31. The animal pad through the failed northern fence is at the 1,050 m mark, 138 m before the first carcass on the same side. Eleven entries in the aerodrome's sighting log record kangaroos on the sealed surface of runway 09/27, all around last light. No inspection had been carried out since 07:35 local that morning.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Virtually certain. Two carcasses on the sealed surface, the first officer's sighting call on the CVR, and the recorded impact at 07:30:33 each establish it on their own. Previously graded 90% to carry doubt about where the animals came through the fence, which is not what this proposition says: the object states that they entered the strip and reached the seal, and that is settled. The route is a separate question and it is carried where it belongs, in the fence object's Influence test.</p> |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Pass, extremely likely (95%)</b>   |
| <p><b>Argument For:</b> Both the reject and the strike follow from this. The first officer's call at 07:30:31 is the only trigger for the reject on the CVR, and the impact at 07:30:33 is with these two animals.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> The reject and the strike both follow directly from the animals being on the surface.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Not tested</b>                     |

## No. 2 propeller and right main gear struck two kangaroos

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| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> Impact at 07:30:33 at 114 kt, one second after the reject began. The FDR records a lateral acceleration spike of 0.31 g right and a vertical spike of 1.48 g. Three No. 2 propeller blades were damaged beyond repairable limits with one blade tip separated, and animal tissue was recovered from two blade roots. The right main gear doors were distorted and the lower right fuselage skin punctured by propeller debris. No. 2 propeller RPM decays from the moment of impact and the No. 2 reverse interlock did not make.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Recorded data, physical damage and recovered tissue all agree.</p> |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> The strike removed one of the two reverse units two seconds into a stop with a 67 m scheduled margin, and it is why the aircraft went off the end rather than stopping on it.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Directly and measurably: the achieved deceleration with one reverse was approximately 15% below the certified accelerate-stop value.</p>                                                                                                                                                                                                                                                                               |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Not tested</b>                     |

## Deceleration below the scheduled accelerate-stop performance

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| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> Scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m, assuming the stop begins at or before V1 with symmetric reverse. Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along the runway at the reject, leaving 695 m of sealed surface, and the stopping distance actually achieved from 118 kt was 757 m. The mean deceleration from maximum braking to the runway end was 2.6 m/s<sup>2</sup>, approximately 15% below the value used in the certified data for this weight and condition, with reverse available on No. 1 only.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Recorded data and the distance reconstruction agree with the physical marks on the runway.</p> |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> With reverse on No. 1 only, the stop decelerated about 15% below the certified value. That is the strike's contribution to the overrun, separate from the late start of the stop, which reaches the critical event by its own arrow. Both contributions are needed to reach the end; neither alone does.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> At the certified deceleration from the same point, the reconstruction puts the stop at roughly 660 m, inside the 695 m of runway remaining; at the deceleration achieved it needed 757 m.</p>                                                                                                                                                                                                        |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Not tested</b>                     |

## Runway overrun onto the runway end safety area

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| <b>Existence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Pass, virtually certain (100%)</b> |
| <p><b>Argument For:</b> Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, departed the end of runway 09 at Tarlton Springs Aerodrome at approximately 21 kt at 07:30:51 UTC on 18/06/2026 and came to rest at 07:30:57, 62 m into the 90 m runway end safety area, upright and on the extended centreline, with the nose wheel in graded gravel. 44 passengers, 2 flight crew and 1 cabin crew were on board. Continuous tyre marks run from 1,160 m to the runway end and the nose wheel rut continues to the stopping point.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Nil</p>                              |                                       |
| <b>Influence:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Not tested</b>                     |
| <p><b>Argument For:</b> Substantial damage: three No. 2 propeller blades beyond limits, a shock-load inspection on the engine, distorted right main gear doors, both right main tyres deflated through the fuse plugs at 84% of certified brake energy, and punctures to the lower right fuselage. Nil serious injuries; one cabin crew member reported bruising during the disembarkation. Two kangaroos killed. The aerodrome closed for 4 hours 20 minutes. The aircraft stopped 28 m short of the end of the prepared safety area; beyond it the terrain is ungraded.</p> <p><b>Argument Against:</b> Nil</p> <p><b>Reasoning:</b> Nil</p> |                                       |
| <b>Importance:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Not tested</b>                     |